- STEP 1: LOOPS (20 exercises)

Goal: flow control + repetition

1. Print numbers 1 → 10
2. Print numbers 10 → 1
3. Print all even numbers till N
4. Print all odd numbers till N
5. Sum of first N natural numbers
6. Sum of even numbers till N
7. Count digits of a number
8. Reverse a number
9. Sum of digits
10. Check palindrome number
11. Factorial (without function)
12. Power of a number (a^b using loop)
13. Print multiplication table of N
14. Find largest digit in a number
15. Check prime number
16. Print all primes till N
17. Fibonacci series till N terms
18. Count how many times a digit appears
19. Armstrong number check
20. Strong number check

- STEP 2: IF–ELSE (20 exercises)

Goal: decision-making clarity

1. Check even or odd
2. Check positive, negative, or zero
3. Find greater of two numbers
4. Find greatest of three numbers
5. Check leap year
6. Check vowel or consonant
7. Check alphabet, digit, or special character
8. Check pass or fail (marks)
9. Grade system (A/B/C/F)
10. Bonus salary calculation

11. Electricity bill calculation
12. Ticket price based on age
13. Login validation (username + password)
14. Triangle validity
15. Triangle type
16. Maximum of four numbers
17. Profit or loss calculation
18. Divisible by 5 and 11
19. Check year is century or not
20. Scholarship eligibility logic

- STEP 3: SWITCH CASE (20 exercises)

Goal: structured branching
1. Simple calculator
2. Day number → day name
3. Month number → month name
4. Weekday or weekend
5. Vowel or consonant
6. Traffic light system
7. Menu-driven calculator
8. Currency converter
9. Grade system
10. Units → electricity bill slab
11. Number → spelling (1–5)
12. Area calculator menu
13. Temperature conversion
14. Login menu system
15. Game menu (start, pause, exit)
16. Student result menu
17. Language selector
18. ATM menu system
19. Shape selection
20. Pattern selection menu

- STEP 4: BREAK & CONTINUE (20 exercises)

Goal: loop control mastery

1. Stop loop at 7
2. Skip multiples of 5
3. First prime number break
4. Stop when input is negative
5. Skip even numbers
6. Password attempts limit
7. Exit loop on correct PIN
8. Skip vowels in string
9. Stop on zero input
10. Skip numbers divisible by 3
11. Infinite loop exit case
12. Break on palindrome
13. Continue on invalid input
14. Break after 3 failures
15. Skip spaces in string
16. Stop factorial at zero
17. Menu-driven loop exit
18. Stop printing at 100
19. Skip prime numbers
20. Exit nested loop using flag

- STEP 5: ARRAYS 1D (20 exercises)

Goal: data handling + traversal

1. Input and print array
2. Sum of elements
3. Average
4. Max element
5. Min element
6. Count even and odd
7. Reverse array
8. Search element
9. Frequency of elements
10. Copy array

11. Sort array (ascending)
12. Sort array (descending)
13. Second largest element
14. Duplicate elements
15. Remove duplicates
16. Rotate array
17. Check palindrome array
18. Merge two arrays
19. Insert element
20. Delete element

- STEP 6: ARRAYS 2D (20 exercises)

Goal: matrix thinking
1. Input & display matrix
2. Sum of all elements
3. Row-wise sum
4. Column-wise sum
5. Diagonal sum
6. Transpose matrix
7. Matrix addition
8. Matrix subtraction
9. Matrix multiplication
10. Check symmetric matrix
11. Check identity matrix
12. Upper triangular
13. Lower triangular
14. Sparse matrix check
15. Largest element
16. Smallest element
17. Sum of boundary elements
18. Interchange diagonals
19. Print zig-zag matrix
20. Check magic matrix